CS 215: Task 09: Due: Nov-05; time as announced on Piazza.

1. Suppose the response follows the simple linear regression model

$$Y = \alpha + \beta x + e,$$

where x is fixed, $\mathbb{E}[e] = 0$, and $0 < \beta < 1$. Show that if

$$x < \frac{\alpha}{1-\beta},$$

then

$$x < \mathbb{E}[Y] < \frac{\alpha}{1-\beta}.$$

Solution: Since x, α, and β are fixed,

$$\mathbb{E}[Y] = \alpha + \beta x + \mathbb{E}[e] = \alpha + \beta x.$$

First, since $1 - \beta > 0$,

$$(1-\beta)x < \alpha.$$

Adding βx to both sides yields

$$x < \alpha + \beta x = \mathbb{E}[Y].$$

For the second inequality, multiply the assumption by $\beta(1-\beta) > 0$ to obtain

$$\beta(1-\beta)x < \beta(1-\beta)\frac{\alpha}{1-\beta} = \alpha\beta.$$

Adding $(1-\beta)\alpha$ to both sides gives

$$(1-\beta)(\alpha + \beta x) < \alpha\beta + (1-\beta)\alpha = \alpha.$$

Dividing by $1 - \beta > 0$,

$$\mathbb{E}[Y] = \alpha + \beta x < \frac{\alpha}{1-\beta}.$$

Combining the two inequalities,

$$\boxed{x < \mathbb{E}[Y] < \frac{\alpha}{1-\beta}.}$$

2. Consider the simple linear regression model

$$Y_i = \alpha + \beta x_i + e_i, \qquad i = 1, \ldots, n,$$

where the predictors are fixed and the errors are independent random variables with mean 0 and variance $\sigma^2 > 0$. Let A and B be the ordinary least squares estimators, and define

$$\widehat{Y}_i = A + Bx_i, \qquad R_i = Y_i - \widehat{Y}_i.$$

Let, as usual, $\bar{x} = n^{-1}\sum_i x_i$ and $S_{xx} = \sum_i (x_i - \bar{x})^2 > 0$.

(a) Derive $\mathrm{Cov}(A, B)$ in terms $\bar{x}$, σ^2 and S_{xx}. Recall that $\mathrm{Var}(B) = \frac{\sigma^2}{S_{xx}}$ as mentioned in class.

Solution: Recall the ordinary least squares estimators:

$$B = \frac{\sum_{i=1}^n (x_i - \bar{x})Y_i}{S_{xx}}, \qquad A = \bar{Y} - B\bar{x}.$$

Since $\bar{x}$ is fixed,

$$\begin{aligned}\mathrm{Cov}(A, B) &= \mathrm{Cov}(\bar{Y} - \bar{x}B, B) \\ &= \mathrm{Cov}(\bar{Y}, B) - \bar{x}\,\mathrm{Var}(B).\end{aligned}$$

We are given

$$\mathrm{Var}(B) = \frac{\sigma^2}{S_{xx}}.$$

Thus, it remains to calculate $\mathrm{Cov}(\bar{Y}, B)$.
Writing $d_i = x_i - \bar{x}$,

$$\bar{Y} = \frac{1}{n}\sum_{j=1}^n Y_j, \qquad B = \frac{1}{S_{xx}}\sum_{i=1}^n d_i Y_i.$$

Using bilinearity of covariance,

$$\mathrm{Cov}(\bar{Y}, B) = \frac{1}{nS_{xx}}\sum_{j=1}^n \sum_{i=1}^n d_i \,\mathrm{Cov}(Y_j, Y_i).$$

Because the predictors are fixed, the randomness in Y_i comes only from e_i. Independence of the errors gives

$$\mathrm{Cov}(Y_j, Y_i) = \begin{cases} \sigma^2, & i = j, \\ 0, & i \neq j. \end{cases}$$

Therefore, only terms with $i = j$ remain:

$$\begin{aligned}\mathrm{Cov}(\bar{Y}, B) &= \frac{\sigma^2}{nS_{xx}}\sum_{i=1}^n d_i \\ &= \frac{\sigma^2}{nS_{xx}}\sum_{i=1}^n (x_i - \bar{x}) \\ &= 0.\end{aligned}$$

The last equality follows from the defining property of the sample mean: $\sum_i (x_i - \bar{x}) = 0$.
Consequently,

$$\boxed{\mathrm{Cov}(A, B) = -\frac{\bar{x}\sigma^2}{S_{xx}}.}$$

Interpretation. Since $A = \bar{Y} - \bar{x}B$, an increase in the estimated slope must be accompanied by an adjustment in the intercept. The sign of this relationship depends on $\bar{x}$.

(b) Under what condition on the predictor values is $\mathrm{Cov}(A, B) = 0$?

Solution: From part (a),

$$\mathrm{Cov}(A, B) = -\frac{\bar{x}\sigma^2}{S_{xx}}.$$

Since $\sigma^2 > 0$ and $S_{xx} > 0$, the covariance vanishes if and only if

$$\boxed{\bar{x} = 0 \iff \sum_{i=1}^{n} x_i = 0.}$$

(c) A student writes on Piazza: "The observation Y_i has variance σ^2. Subtracting the estimated fitted value $\widehat{Y}_i$ adds uncertainty" Dissect the student's argument explaining where the student is right. Is the intended conclusion "R_i must have variance at least σ^2" true or false?
Also explain intuitively your answer.

Solution: The student's observation that $\widehat{Y}_i$ is random is correct: the fitted value depends on the estimated coefficients A and B.

However, the conclusion is **false**.

The key mistake is implicitly treating Y_i and $\widehat{Y}_i$ as independent or uncorrelated.

In general,

$$\mathrm{Var}(Y_i - \widehat{Y}_i) = \mathrm{Var}(Y_i) + \mathrm{Var}(\widehat{Y}_i) - 2\,\mathrm{Cov}(Y_i, \widehat{Y}_i).$$

The student correctly identifies the first two terms, but ignores the covariance term.

Intuition. The fitted value $\widehat{Y}_i$ is not an independent noisy quantity: it is constructed using the same observations, including Y_i.

In fact, it can be shown that

$$\mathrm{Var}(R_i) = \sigma^2 \left(1 - \frac{1}{n} - \frac{(x_i - \bar{x})^2}{S_{xx}}\right).$$

so the variance is actually smaller in general.

3. Suppose the observations follow the simple linear regression model

$$Y_i = \alpha + \beta x_i + e_i, \qquad i = 1, \ldots, n,$$

where the predictors are fixed, $S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2 > 0$, and the errors are independent $N(0, \sigma^2)$ random variables. Consider the alternative parameterization

$$V_i = \gamma + \beta(x_i - \bar{x}) + e_i, \qquad \gamma = \alpha + \beta\bar{x}. \tag{1}$$

Let (A, B) and (C, B^*) denote the maximum likelihood estimators under the original and alternative parameterizations, respectively. Here C estimates γ, and B^* estimates β in Eq. 1. The coefficient estimators maximize the Gaussian likelihood with respect to the coefficients for each fixed $\sigma^2 > 0$.

(a) Reason the equivalence of V_i and Y_i and thus $B^* = B$. Also mathematically show this.

(b) Show that $C = \bar{Y}$ and derive its distribution.

Solution:

(a) **Equivalence and equality of slope estimators.**

The alternative model is obtained by centering the predictor around its sample mean.

Indeed, using $\gamma = \alpha + \beta\bar{x}$,

$$\begin{aligned} V_i &= \gamma + \beta(x_i - \bar{x}) + e_i \\ &= \alpha + \beta\bar{x} + \beta x_i - \beta\bar{x} + e_i \\ &= \alpha + \beta x_i + e_i \\ &= Y_i. \end{aligned}$$

Thus, the two parameterizations describe exactly the same observations and fitted regression line.

Only the interpretation of the intercept changes: α is the fitted mean at $x = 0$, while γ is the fitted mean at $x = \bar{x}$.

Since shifting the origin of the predictor does not change its relationship with the response, the estimated slope should remain unchanged.

To show this mathematically, let

$$z_i = x_i - \bar{x}.$$

Then $\bar{z} = 0$, and the centered-model residual sum of squares is

$$Q(\gamma, \beta) = \sum_{i=1}^{n}(Y_i - \gamma - \beta z_i)^2.$$

Differentiating with respect to β,

$$\frac{\partial Q}{\partial \beta} = -2\sum_{i=1}^{n} z_i(Y_i - \gamma - \beta z_i).$$

At the minimum,

$$\sum_i z_i Y_i - \gamma\sum_i z_i - \beta\sum_i z_i^2 = 0.$$

Because $\sum_i z_i = 0$,

$$B^* = \frac{\sum_i z_i Y_i}{\sum_i z_i^2} = \frac{\sum_i (x_i - \bar{x})Y_i}{S_{xx}}.$$

For the original model, the OLS slope estimator is

$$B = \frac{\sum_i (x_i - \bar{x})(Y_i - \bar{Y})}{S_{xx}}.$$

But

$$\sum_i (x_i - \bar{x})\bar{Y} = \bar{Y} \sum_i (x_i - \bar{x}) = 0.$$

Therefore,

$$\boxed{B^* = B.}$$

(b) **Estimator and distribution of C.**

To find the intercept estimator, differentiate the centered-model residual sum of squares with respect to γ:

$$\frac{\partial Q}{\partial \gamma} = -2 \sum_{i=1}^{n} (Y_i - \gamma - \beta z_i).$$

Setting this derivative equal to zero at the optimum,

$$nC = \sum_{i=1}^{n} Y_i - B^* \sum_{i=1}^{n} z_i.$$

Since the centered predictors sum to zero,

$$\boxed{C = \bar{Y}.}$$

To derive its distribution, average the original model:

$$\begin{aligned} \bar{Y} &= \frac{1}{n} \sum_{i=1}^{n} (\alpha + \beta x_i + e_i) \\ &= \alpha + \beta \bar{x} + \bar{e} \\ &= \gamma + \bar{e}, \end{aligned}$$

where

$$\bar{e} = \frac{1}{n} \sum_{i=1}^{n} e_i.$$

Since the errors are independent normal random variables, their average is also normally distributed, with

$$\mathbb{E}[\bar{e}] = 0, \qquad \mathrm{Var}(\bar{e}) = \frac{1}{n^2} \sum_{i=1}^{n} \sigma^2 = \frac{\sigma^2}{n}.$$

Therefore,

$$\boxed{C = \bar{Y} \sim N\left(\gamma, \frac{\sigma^2}{n}\right).}$$

4. The response random variables $Y_1, \ldots, Y_n$ follow the model

$$Y_i = \theta x_i^2 + e_i, \qquad i = 1, \ldots, n,$$

where $x_1, \ldots, x_n$ are fixed known constants, not all zero, and $e_1, \ldots, e_n$ are independent $N(0, \sigma^2)$ random variables, with known $\sigma^2 > 0$. The parameter $\theta \in \mathbb{R}$ is unknown.

(a) Find the least squares estimator of θ.

(b) Derive the maximum likelihood estimator of θ.

(c) Which estimator from that obtained in (a) and (b) is unbiased? Find its variance.

Solution:

(a) **Least squares.** For observed responses $y_1, \ldots, y_n$, minimize

$$Q(\theta) = \sum_{i=1}^{n} (y_i - \theta x_i^2)^2.$$

Differentiating gives

$$Q'(\theta) = -2 \sum_i x_i^2 y_i + 2\theta D, \qquad Q''(\theta) = 2D > 0.$$

where $D = \sum_{i=1}^n x_i^4$ assumed to be not equal to 0.

Thus the unique least squares estimate and its corresponding estimator are

$$\boxed{t_{\mathrm{LS}} = \frac{\sum_i x_i^2 y_i}{D}, \qquad T_{\mathrm{LS}} = \frac{\sum_i x_i^2 Y_i}{D}.}$$

(b) **Maximum likelihood.**

With the errors are independent normal random variables with mean zero and known variance $\sigma^2 > 0$, the observed-data likelihood as a function of θ is (the notation for the likelihood is slightly different from the earlier task)

$$L(\theta; y_1, \ldots, y_n) = (2\pi\sigma^2)^{-n/2} \exp\left\{ -\frac{1}{2\sigma^2} \sum_i (y_i - \theta x_i^2)^2 \right\}.$$

Its logarithm is

$$\ell(\theta; y_1, \ldots, y_n) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{Q(\theta)}{2\sigma^2}.$$

Maximizing this is equivalent to minimizing Q. Consequently,

$$t_{\mathrm{ML}} = t_{\mathrm{LS}}, \qquad \boxed{T_{\mathrm{ML}} = T_{\mathrm{LS}} = \frac{\sum_i x_i^2 Y_i}{D}.}$$

(c) **Expectation and variance.** Denote the common estimator by T. Substitution of the model gives

$$T = \frac{\sum_i x_i^2 (\theta x_i^2 + e_i)}{D} = \theta + \frac{\sum_i x_i^2 e_i}{D}.$$

Hence

$$\mathbb{E}_\theta[T] = \theta, \qquad \mathrm{Var}_\theta(T) = \frac{\sigma^2 \sum_i x_i^4}{D^2} = \frac{\sigma^2}{D}.$$

Therefore the common estimator is unbiased, with

$$\boxed{T = \frac{\sum_i x_i^2 Y_i}{\sum_i x_i^4}, \qquad \mathrm{Var}_\theta(T) = \frac{\sigma^2}{\sum_i x_i^4}.}$$

Interpretation. Defining $z_i = x_i^2$ implies $Y_i = \theta z_i + e_i$, a straight-line regression through the origin in the transformed predictor z_i. The model is linear in the unknown parameter θ, even though the mean response is quadratic in the original input x_i.

5. This is a computational exercise. Use Python, the numpy and matplot packages. We estimate about 10-20 lines of python code.

   Ten experiments on wood samples result in the following data. The input x_i is specific gravity and y_i is the observed maximum crushing strength parallel to the grain, measured in psi.

i	x_i	y_i	i	x_i	y_i
1	.41	1850	6	.39	1760
2	.46	2620	7	.41	2500
3	.44	2340	8	.44	2750
4	.47	2690	9	.43	2730
5	.42	2160	10	.44	3120

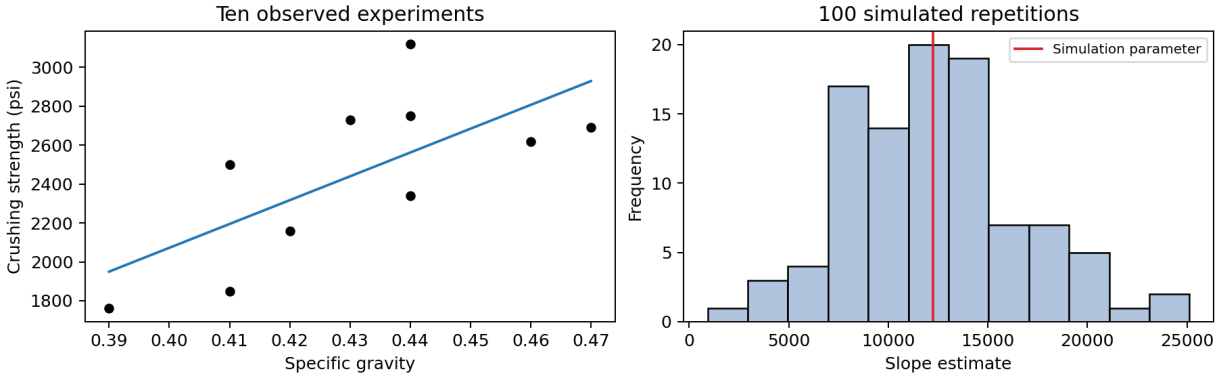
   Consider $Y_i = \alpha + \beta x_i + e_i$, where the inputs are fixed and the errors are independent $N(0, \sigma^2)$ random variables. We write A, B for the least squares estimators and a, b for their observed values.

   (a) Draw a scatter plot and fit the regression line. See the sample shown.

   (b) What is a, b and the estimated mean strength at $x = .43$ (Why this number?)

   (c) Estimate the variance.

   Use the fitted values a, b, s^2 as the fixed parameters of a simulation model. Perform exactly $M = 100$ independent repetitions of the ten-experiment design. Use the random number generator seed as 215 (Why?). Do not resample or change the input levels.



   (d) Compute the empirical means and sample variances of the 100 intercept and slope estimates.

   (e) Compare them with the theoretical means and variances under this simulation model.

   (f) Plot the simulated slope estimates.

Solution:

(a) With $n = 10$, the summary quantities are

$$\bar{x} = .431, \qquad \bar{y} = 2452, \qquad S_{xx} = .00529.$$

Estimators are $B = S_{xY}/S_{xx}$ and $A = \bar{Y} - B\bar{x}$. Their observed values are

$$b \approx 12245.75, \qquad a \approx -2825.92, \qquad \widehat{m}(.43) = a + .43b \approx 2439.75 \text{ psi}.$$

(b) The residuals, in table order $i = 1, \ldots, 10$, are approximately

$$(r_1, \ldots, r_5) \approx (-344.839, -187.127, -222.212, -239.584, -157.297),$$
$$(r_6, \ldots, r_{10}) \approx (-189.924, 305.161, 187.788, 290.246, 557.788).$$

Direct computation results in

$$\text{ssr} \approx 845280.5293 \text{ psi}^2, \qquad s^2 \approx 105660.07 \text{ psi}^2.$$

Here s^2 is an observed value of the unbiased estimator $S^2 = \text{SSR}/(n-2)$ of the common error variance σ^2.

(c) Denote the simulation parameters by $\alpha_0 = a$, $\beta_0 = b$, and $\sigma_0^2 = s^2$. These are fixed numbers in the simulation, not the unknown true parameters of the original population.

In each repetition the estimators satisfy

$$\mathbb{E}[A] = \alpha_0 \approx -2825.92, \quad \text{Var}(A) = \sigma_0^2 \left(\frac{1}{10} + \frac{\bar{x}^2}{S_{xx}} \right),$$

$$\mathbb{E}[B] = \beta_0 \approx 12245.75, \quad \text{Var}(B) = \frac{\sigma_0^2}{S_{xx}}.$$

The theoretical variances are approximately 3.72×10^6 and 1.99×10^7, respectively.

With the code below the empirical means are approximately -2891.90 and 12388.95, and the empirical variances are 3.58×10^6 and 1.93×10^7.

The following code implements the experiment;

`ddof=1` gives the sample variances across repetitions.

```
import numpy as np
import matplotlib.pyplot as plt
x = np.array([.41,.46,.44,.47,.42,.39,.41,.44,.43,.44])
y = np.array([1850,2620,2340,2690,2160,1760,2500,2750,2730,3120])
X = np.column_stack((np.ones(x.size), x))
a, b = np.linalg.lstsq(X, y, rcond=None)[0]
r = y - X @ np.array([a, b])
s2 = (r @ r) / (x.size - 2)
rng = np.random.default_rng(215)
ys = a + b*x[:, None] + rng.normal(0, np.sqrt(s2), (10, 100))
coef = np.linalg.lstsq(X, ys, rcond=None)[0]
print(coef.mean(axis=1))
print(coef.var(axis=1, ddof=1))
plt.hist(coef[1], bins=12, edgecolor="black")
plt.axvline(b, color="red", label="Simulation slope parameter")
plt.xlabel("Observed slope estimate in one repetition")
plt.ylabel("Frequency")
plt.legend()
plt.show()
```

6. **Multiple regression: investigate the computation.** This is a computational exercise; Python packages may be used. Ten observations of four predictors and a response are provided below. Fit the model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + e_i,$$

with fixed inputs and independent $N(0, \sigma^2)$ errors. Include an intercept. Denote the design matrix by X and the observed response vector by $\mathbf{y}$.

x_1	x_2	x_3	x_4	y
0	0	5	7	97
16	4	1	11	275
9	-3	8	4	246
4	2	3	9	140
25	-5	10	2	482
1	-1	6	6	122
9	3	2	10	183
16	-4	9	3	359
1	1	4	8	82
4	-2	7	5	146

(a) First attempt to compute the coefficient estimates by solving

$$(X^\mathsf{T} X)\mathbf{b} = X^\mathsf{T}\mathbf{y}$$

using `numpy.linalg.solve`. Record any error or warning. If values are returned, retain them for investigation rather than assuming success.

(b) Obtain an equivalent model with a full-column-rank matrix and fit it. Exhibit two distinct coefficient vectors for the original model that results in the same fitted responses.

Solution:

(a) In the tested NumPy environment, solving the normal equations raises a `LinAlgError` with message "Singular matrix."

(b) For every observation,

$$x_{i2} + x_{i3} = 5, \qquad x_{i3} + x_{i4} = 12.$$

Consequently, the five columns of X span only the three-dimensional space generated by $\mathbf{1}$, the x_1 column, and the x_3 column. These three columns are independent, so $\operatorname{rank}(X) = 3$. Thus $X^\mathsf{T} X$ is singular and the original coefficient vector is not uniquely determined. A pseudoinverse-based solver such as `numpy.linalg.lstsq` can select one solution (the minimum Euclidean norm solution), but it cannot make the original parameters identifiable.

Multiple solutions are therefore possible. Here is one.

Substituting $x_2 = 5 - x_3$ and $x_4 = 12 - x_3$ results in

$$\mathbb{E}[Y_i] = \gamma_0 + \gamma_1 x_{i1} + \gamma_3 x_{i3},$$

where

$$\gamma_0 = \beta_0 + 5\beta_2 + 12\beta_4, \qquad \gamma_1 = \beta_1, \qquad \gamma_3 = \beta_3 - \beta_2 - \beta_4.$$

These three combinations are identifiable. Their observed least squares estimates result in

$$\widehat{m}(x_1, x_3) = 43.32 + 13.94_1 + 9.36_3.$$

In the order $(\beta_0, \beta_1, \beta_2, \beta_3, \beta_4)$, two possible coefficient estimates are

$$\mathbf{b}^{(1)} = (43.32,\ 13.94,\ 0,\ 9.36,\ 0)^{\mathsf{T}},$$
$$\mathbf{b}^{(2)} = (38.32,\ 13.94,\ 1,\ 10.35,\ 0)^{\mathsf{T}}.$$